

Supplementary Appendix: Difference-in-differences with “bad controls”

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This supplementary appendix contains proofs for some of the results in the main text as well as additional results and supporting details for the arguments in the main text. Appendix SA proves asymptotic normality of the imputation estimator provided in Section 6.1 in the main text. Appendix SB proves double robustness of the Neyman-orthogonal estimator proposed in Section 6.2 and then proves asymptotic normality when nuisance functions are estimated by machine learning. Appendix SC develops an alternative identification strategy under parallel trends for the bad control. Finally, Appendix SD provides Monte Carlo simulations.

SA Proofs for Imputation Estimator

The following assumption collects standard regularity conditions for Proposition 4. Define the vectors of regressors $R_i := (X_{it^*}, X_{i,t^*-1}, Z_i)'$ and $S_i := (X_{i,t^*-1}, W_i', Z_i)'$.

Assumption S1 (Regularity Conditions for Imputation Estimator).

- (i) $\mathbb{E}\|X_{t^*}\|^4$, $\mathbb{E}\|X_{t^*-1}\|^4$, $\mathbb{E}\|W\|^4$, $\mathbb{E}\|Z\|^4$, and $\mathbb{E}[\Delta Y_{it^*}^4]$ are finite.
- (ii) The matrices $\mathbb{E}[RR' \mid D = 0]$ and $\mathbb{E}[SS' \mid D = 0]$ are positive definite.

Throughout this section, we use notation from Section 6.1 and write $\pi = P(D = 1)$ and $n_0 = \sum_i (1 - D_i)$. Under Assumption 8, we have $m_0(X_{t^*}, X_{t^*-1}, Z) = R' \beta$, $\mathbb{E}[X_{t^*} \mid S, D = 0] = S' \gamma$, and $\nu_0(X_{t^*-1}, W, Z) = S' \gamma \beta_1 + X_{t^*-1} \beta_2 + Z' \beta_3 = \tilde{R}' \beta$. The OLS estimators use only untreated observations:

$$\begin{aligned}\hat{\beta} &= \left(\sum_{i=1}^n (1 - D_i) R_i R_i' \right)^{-1} \sum_{i=1}^n (1 - D_i) R_i \Delta Y_{it^*}, \\ \hat{\gamma} &= \left(\sum_{i=1}^n (1 - D_i) S_i S_i' \right)^{-1} \sum_{i=1}^n (1 - D_i) S_i X_{it^*}.\end{aligned}$$

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Define the error $u_i = \Delta Y_{it^*} - R'_i \beta$ and $v_i = X_{it^*} - S'_i \gamma$, which satisfy $\mathbb{E}[u \mid R, D = 0] = 0$ and $\mathbb{E}[v \mid S, D = 0] = 0$ under Assumption 8. Let $\tilde{R}_i = ((S'_i \gamma)', X'_{i,t^*-1}, Z'_i)'$ correspond to R_i above but with X_{it^*} replaced by its conditional mean $S'_i \gamma$, and let

$$\Sigma_R = \mathbb{E} \left[\frac{1-D}{1-\pi} R R' \right], \quad \Sigma_S = \mathbb{E} \left[\frac{1-D}{1-\pi} S S' \right].$$

The following lemma provides the asymptotic linear representations for $\hat{\beta}$ and $\hat{\gamma}$, which are standard consequences of OLS theory.

Lemma S1. *Under Assumption S1,*

$$\sqrt{n}(\hat{\beta} - \beta) = \Sigma_R^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{1-D_i}{1-\pi} R_i u_i + o_p(1), \quad (\text{S1})$$

$$\sqrt{n}(\hat{\gamma} - \gamma) = \Sigma_S^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{1-D_i}{1-\pi} S_i v_i + o_p(1). \quad (\text{S2})$$

Proof. For $\sqrt{n}(\hat{\beta} - \beta)$, notice that

$$\begin{aligned} \sqrt{n}(\hat{\beta} - \beta) &= \left(\frac{1}{n} \sum_{i=1}^n \frac{1-D_i}{1-\hat{\pi}} R_i R'_i \right)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{1-D_i}{1-\hat{\pi}} R_i u_i \\ &= \left(\frac{1}{n} \sum_{i=1}^n \frac{1-D_i}{1-\pi} R_i R'_i \right)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{1-D_i}{1-\pi} R_i u_i \\ &= \Sigma_R^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{1-D_i}{1-\pi} R_i u_i + o_p(1) \end{aligned}$$

where the first equality holds by the definition of $\hat{\beta}$, the second equality holds by canceling the $(1-\hat{\pi})$ terms and then multiplying and dividing by $(1-\pi)$, and the last equality holds by the law of large numbers applied to the first term. An analogous argument holds for $\sqrt{n}(\hat{\gamma} - \gamma)$. $\square$

Proof of Proposition 4. First, we decompose

$$\sqrt{n}(\widehat{\text{ATT}}_{ra} - \text{ATT}) = \underbrace{\sqrt{n}(\hat{m}_1 - \mathbb{E}[\Delta Y_{t^*} \mid D = 1])}_A - \underbrace{\sqrt{n}(\hat{\tau}_{ra} - \mathbb{E}[\nu_0 \mid D = 1])}_B. \quad (\text{S3})$$

For term A in Equation (S3),

$$\begin{aligned} \sqrt{n}(\hat{m}_1 - \mathbb{E}[\Delta Y_{t^*} \mid D = 1]) &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{D_i}{\hat{\pi}} \left(\Delta Y_{it^*} - \mathbb{E}[\Delta Y_{t^*} \mid D = 1] \right) \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{D_i}{\pi} \left(\Delta Y_{it^*} - \mathbb{E}[\Delta Y_{t^*} \mid D = 1] \right) + o_p(1) \end{aligned} \quad (\text{S4})$$

where the first equality holds by the definition of $\hat{m}_1$, the second equality holds by multiplying and dividing by π and because $\pi/\hat{\pi} \xrightarrow{p} 1$.

For term B in Equation (S3), from Equation (7) in the main text, we have that

$$\hat{\tau}_{ra} = \frac{1}{n} \sum_{i=1}^n \frac{D_i}{\hat{\pi}} \hat{\nu}_{0,i} \quad \text{where } \hat{\nu}_{0,i} := \hat{\nu}_0(X_{t^*-1}, W, Z) = S' \hat{\gamma} \hat{\beta}_1 + X_{t^*-1} \hat{\beta}_2 + Z' \hat{\beta}_3,$$

and we also use the shorthand notation $\nu_{0,i} := \nu_0(X_{i,t^*-1}, W_i, Z_i)$. Next, notice that

$$\sqrt{n} \left(\hat{\tau}_{ra} - \mathbb{E}[\nu_0 \mid D = 1] \right) = \underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{D_i}{\hat{\pi}} (\hat{\nu}_{0,i} - \nu_{0,i})}_{B_1} + \underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{D_i}{\hat{\pi}} (\nu_{0,i} - \mathbb{E}[\nu_0 \mid D = 1])}_{B_2},$$

For B_1 , notice that

$$\begin{aligned} \hat{\nu}_{0,i} - \nu_{0,i} &= \hat{\nu}_{0,i} - S'_i \gamma \hat{\beta}_1 + S'_i \gamma \hat{\beta}_1 - \nu_{0,i} \\ &= S'_i (\hat{\gamma} - \gamma) \hat{\beta}_1 + \tilde{R}'_i (\hat{\beta} - \beta). \end{aligned} \tag{S5}$$

which implies that

$$\begin{aligned} \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{D_i}{\hat{\pi}} (\hat{\nu}_{0,i} - \nu_{0,i}) &= \left(\frac{1}{n} \sum_{i=1}^n \frac{D_i}{\pi} \beta_1 S'_i \right) \sqrt{n} (\hat{\gamma} - \gamma) + \left(\frac{1}{n} \sum_{i=1}^n \frac{D_i}{\pi} \tilde{R}'_i \right) \sqrt{n} (\hat{\beta} - \beta) + o_p(1) \\ &= \mathbb{E} \left[\frac{D}{\pi} \beta_1 S'_i \right] \sqrt{n} (\hat{\gamma} - \gamma) + \mathbb{E} \left[\frac{D}{\pi} \tilde{R}' \right] \sqrt{n} (\hat{\beta} - \beta) + o_p(1) \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{1 - D_i}{1 - \pi} \left(\mathbb{E} \left[\frac{D}{\pi} \beta_1 S' \right] \Sigma_S^{-1} S_i v_i + \mathbb{E} \left[\frac{D}{\pi} \tilde{R} \right]' \Sigma_R^{-1} R_i u_i \right) + o_p(1). \end{aligned} \tag{S6}$$

where the first equality holds by plugging in Equation (S5) and by consistency of $\hat{\pi}$ and $\hat{\beta}_1$, the second equality holds by applying the law of large numbers and continuous mapping theorem to the two sample average terms in the previous line, and the last equality holds by plugging in the result in Lemma S1.

For B_2 , it holds immediately that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{D_i}{\hat{\pi}} (\nu_{0,i} - \mathbb{E}[\nu_0 \mid D = 1]) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{D_i}{\pi} (\nu_{0,i} - \mathbb{E}[\nu_0 \mid D = 1]) + o_p(1) \tag{S7}$$

because $\hat{\pi}$ is consistent for π and by the continuous mapping theorem.

Next,

$$\begin{aligned}\psi_i^{ra} &:= \frac{D_i}{\pi} \left(\Delta Y_{it^*} - \mathbb{E}[\Delta Y_{t^*} \mid D = 1] - \nu_{0,i} + \mathbb{E}[\nu_0 \mid D = 1] \right) \\ &\quad - \frac{1 - D_i}{1 - \pi} \left(\mathbb{E} \left[\frac{D}{\pi} \tilde{R} \right]' \Sigma_R^{-1} R_i u_i + \mathbb{E} \left[\frac{D}{\pi} \beta_1' S' \right] \Sigma_S^{-1} S_i v_i \right).\end{aligned}\tag{S8}$$

Then, by combining Equations (S4), (S6) and (S7), it follows that

$$\sqrt{n}(\widehat{\text{ATT}}_{ra} - \text{ATT}) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_i^{ra} + o_p(1) \xrightarrow{d} \mathcal{N}(0, \text{Var}(\psi^{ra}))$$

□

SB Proofs for Neyman Orthogonal Estimator

SB.1 Neyman Orthogonal Estimand

Proof of Proposition 5. We consider the expectation of each line in Equation (10) in the main text in turn.

$$\mathbb{E} \left[\frac{D}{\pi} \Delta Y_{t^*} - \frac{D}{\pi} \nu_0(X_{t^*-1}, W, Z) \right] = \mathbb{E}[\Delta Y_{t^*} \mid D = 1] - \mathbb{E}[\nu_0(X_{t^*-1}, W, Z) \mid D = 1] = \text{ATT},$$

which holds by the law of iterated expectations, Theorem 2, and the definition of ν_0 . Next,

$$\begin{aligned}\mathbb{E} &\left[-\frac{1-D}{\pi} (m_0(X_{t^*}, X_{t^*-1}, Z) - \nu_0(X_{t^*-1}, W, Z)) \frac{p(X_{t^*-1}, W, Z)}{1 - p(X_{t^*-1}, W, Z)} \right] \\ &= -\frac{1-\pi}{\pi} \mathbb{E} \left[m_0(X_{t^*}, X_{t^*-1}, Z) \frac{p(X_{t^*-1}, W, Z)}{1 - p(X_{t^*-1}, W, Z)} \mid D = 0 \right] \\ &\quad + \frac{1-\pi}{\pi} \mathbb{E} \left[\nu_0(X_{t^*-1}, W, Z) \frac{p(X_{t^*-1}, W, Z)}{1 - p(X_{t^*-1}, W, Z)} \mid D = 0 \right] \\ &= 0,\end{aligned}$$

where the first equality holds by the law of iterated expectations, and the second equality holds by applying the law of iterated expectations again to the first term and by the definition of ν_0 . Next,

$$\begin{aligned}\mathbb{E} &\left[-\frac{1-D}{\pi} (\Delta Y_{t^*} - m_0(X_{t^*}, X_{t^*-1}, Z)) \omega_0(X_{t^*}, X_{t^*-1}, Z) \right] \\ &= -\frac{1-\pi}{\pi} \mathbb{E} \left[(\Delta Y_{t^*} - m_0(X_{t^*}, X_{t^*-1}, Z)) \omega_0(X_{t^*}, X_{t^*-1}, Z) \mid D = 0 \right] \\ &= 0\end{aligned}$$

where the first equality holds by the law of iterated expectations, and the second equality holds by applying the law of iterated expectations again and by the definition of m_0 . Combining the previous three displays implies the result. $\square$

Lemma S2. *Under Assumptions 6 and 7, for any integrable function $h(X_{t^*}(0), X_{t^*-1}, Z)$,*

$$\mathbb{E}[h(X_{t^*}(0), X_{t^*-1}, Z) \omega_0(X_{t^*}(0), X_{t^*-1}, Z) \mid D = 0] = \frac{\pi}{1 - \pi} \mathbb{E}[h(X_{t^*}(0), X_{t^*-1}, Z) \mid D = 1].$$

Proof. Notice that, for $h := h(X_{t^*}(0), X_{t^*-1}, Z)$, we have that

$$\begin{aligned} \mathbb{E}[h \omega_0(X_{t^*}(0), X_{t^*-1}, Z) \mid D = 0] &= \mathbb{E}\left[h \frac{p(X_{t^*-1}, W, Z)}{1 - p(X_{t^*-1}, W, Z)} \mid D = 0\right] \\ &= \mathbb{E}\left[\frac{p(X_{t^*-1}, W, Z)}{1 - p(X_{t^*-1}, W, Z)} \mathbb{E}[h \mid X_{t^*-1}, W, Z, D = 0] \mid D = 0\right] \\ &= \mathbb{E}\left[\frac{p(X_{t^*-1}, W, Z)}{1 - p(X_{t^*-1}, W, Z)} \mathbb{E}[h \mid X_{t^*-1}, W, Z, D = 1] \mid D = 0\right] \\ &= \frac{\pi}{1 - \pi} \mathbb{E}[\mathbb{E}[h \mid X_{t^*-1}, W, Z, D = 1] \mid D = 1] \\ &= \frac{\pi}{1 - \pi} \mathbb{E}[h \mid D = 1] \end{aligned}$$

where the first equality holds by the definition of ω_0 in Equation (9) in the main text and the law of iterated expectations, the second equality holds by the law of iterated expectations, the third equality holds by Assumption 6, the fourth equality holds by a change of measure argument, and the last equality holds by the law of iterated expectations. $\square$

Proof of Proposition 6. Since $\widehat{\text{ATT}}_{dr}(\hat{\theta}) \xrightarrow{p} \text{ATT}_{dr}(\theta)$ by assumption, it suffices to show $\text{ATT}_{dr}(\theta) = \text{ATT}$ under each condition, where

$$\text{ATT}_{dr}(\theta) = m_1 - \tau(\theta) - T_3(\theta) - T_4(\theta), \tag{S9}$$

with $m_1 = \mathbb{E}[\Delta Y_{t^*} \mid D = 1]$, $\tau(\theta) = \mathbb{E}[\nu_0(\theta_\nu) \mid D = 1]$,

$$\begin{aligned} T_3(\theta) &= \frac{1 - \pi}{\pi} \mathbb{E}\left[(m_0(\theta_m) - \nu_0(\theta_\nu)) \frac{p(\theta_p)}{1 - p(\theta_p)} \mid D = 0\right], \\ T_4(\theta) &= \frac{1 - \pi}{\pi} \mathbb{E}[(\Delta Y_{t^*} - m_0(\theta_m)) \omega_0(\theta_\omega) \mid D = 0]. \end{aligned}$$

Since $\text{ATT} = m_1 - \tau$ by Theorem 2, we aim to show that $\text{ATT}_{dr}(\theta) = m_1 - \tau$.

Case (i): $(m_0(\theta_m), \nu_0(\theta_\nu)) = (m_0, \nu_0)$.

In this case, $\tau(\theta) = \tau$. For $T_3(\theta)$, notice that

$$\frac{1-\pi}{\pi} \mathbb{E} \left[(m_0 - \nu_0) \frac{p(\theta_p)}{1-p(\theta_p)} \mid D=0 \right] = \frac{1-\pi}{\pi} \mathbb{E} \left[\frac{p(\theta_p)}{1-p(\theta_p)} \underbrace{\mathbb{E}[(m_0 - \nu_0) \mid X_{t^*-1}, W, Z, D=0]}_{=0} \mid D=0 \right] = 0$$

where the first equality holds by the law of iterated expectations and because $p(\theta_p)$ is a deterministic function given (X_{t^*-1}, W, Z) , and the underlined term is equal to 0 by the definition of ν_0 in Equation (5). Next, for $T_4(\theta)$, notice that

$$\frac{1-\pi}{\pi} \mathbb{E} \left[(\Delta Y_{t^*} - m_0) \omega_0(\theta_\omega) \mid D=0 \right] = \frac{1-\pi}{\pi} \mathbb{E} \left[\omega_0(\theta_\omega) \underbrace{\mathbb{E}[(\Delta Y_{t^*} - m_0) \mid X_{t^*}, X_{t^*-1}, Z, D=0]}_{=0} \mid D=0 \right] = 0,$$

where the first equation holds by the law of iterated expectations and because $\omega_0(\theta_\omega)$ is a deterministic function given (X_{t^*}, X_{t^*-1}, Z) , and the underlined term is equal to 0 by the definition of m_0 in Equation (1).

Finally, since $\tau(\theta) = \tau$ and $T_3(\theta) = T_4(\theta) = 0$, it implies that $\text{ATT}_{dr}(\theta) = \text{ATT}$.

Case (ii): $(p(\theta_p), \omega_0(\theta_\omega)) = (p, \omega_0)$.

For $T_3(\theta)$, write

$$T_3(\theta) = \frac{1-\pi}{\pi} \mathbb{E} \left[m_0(\theta_m) \frac{p}{1-p} \mid D=0 \right] - \frac{1-\pi}{\pi} \mathbb{E} \left[\nu_0(\theta_\nu) \frac{p}{1-p} \mid D=0 \right] =: T_3^A(\theta) - T_3^B(\theta),$$

which uses that $p(\theta_p) = p$, and we consider each of these terms individually.

$$T_3^A(\theta) = \frac{1-\pi}{\pi} \mathbb{E} [m_0(\theta_m) \omega_0 \mid D=0] = \mathbb{E} [m_0(\theta_m) \mid D=1] \quad (\text{S10})$$

where the first equality holds by the law of iterated expectations, the definition of ω_0 in Equation (9), and because $m_0(\theta_m)$ is a deterministic function given (X_{t^*}, X_{t^*-1}, Z) , and the second equality holds by Lemma S2. Next,

$$T_3^B(\theta) = \mathbb{E} [\nu_0(\theta_\nu) \mid D=1] = \tau(\theta). \quad (\text{S11})$$

where the first equality holds by a change of measure argument, and the second equality holds by the definition of $\tau(\theta)$. Next, decompose $T_4(\theta)$ as

$$T_4(\theta) = \frac{1-\pi}{\pi} \mathbb{E} [\Delta Y_{t^*} \omega_0 \mid D=0] - \frac{1-\pi}{\pi} \mathbb{E} [m_0(\theta_m) \omega_0 \mid D=0] =: T_4^A(\theta) - T_4^B(\theta),$$

which uses that $\omega_0(\theta_\omega) = \omega_0$. Notice that

$$T_4^A(\theta) = \frac{1-\pi}{\pi} \mathbb{E} \left[\mathbb{E} [\Delta Y_{t^*} \mid X_{t^*}, X_{t^*-1}, Z, D=0] \omega_0 \mid D=0 \right]$$

$$= \frac{1-\pi}{\pi} \mathbb{E}[m_0 \omega_0 \mid D = 0] = \mathbb{E}[m_0 \mid D = 1] = \tau, \quad (\text{S12})$$

where the first equality holds by the law of iterated expectations and because ω_0 is fully determined given (X_{t^*}, X_{t^*-1}, Z) , the second equality holds by the definition of m_0 , the third equality holds by Lemma S2, and the last equality holds by the definition of τ . Next, it immediately follows from Lemma S2 that

$$T_4^B(\theta) = \mathbb{E}[m_0(\theta_m) \mid D = 1] \quad (\text{S13})$$

Combining Equations (S9) to (S13) implies the result. $\square$

SB.2 Asymptotic Normality

The following assumption collects the regularity conditions needed for Proposition 7.

Assumption S2 (Regularity Conditions for DR Estimator).

- (i) $\mathbb{E}[\varphi(O; \eta)^2] < \infty$ and $\text{Var}(\varphi(O; \eta)) > 0$.
- (ii) $\|\hat{m}_0^{-k} - m_0\|_2 = o_p(1)$, $\|\hat{\nu}_0^{-k} - \nu_0\|_2 = o_p(1)$, $\|\hat{p}^{-k} - p\|_2 = o_p(1)$, and $\|\hat{\omega}_0^{-k} - \omega_0\|_2 = o_p(1)$ for each fold k .
- (iii) $p(X_{t^*-1}, W, Z)$ is uniformly bounded away from 1 and $\hat{p}^{-k}(X_{t^*-1}, W, Z)$ is uniformly bounded away from 1 for each fold k .
- (iv) $\hat{\omega}_0^{-k}$ is uniformly bounded for each fold k .
- (v) $\mathbb{E}[\Delta Y_{t^*}^2 \mid X_{t^*}, X_{t^*-1}, Z, D = 0]$ is uniformly bounded.

Proof of Proposition 7. To start with, notice that $\widehat{\text{ATT}}_{dr} = \frac{1}{n} \sum_{i=1}^n \frac{\pi}{\hat{\pi}} \varphi_1(O_i, \hat{\eta}^{-k})$. Then,

$$\begin{aligned} \sqrt{n}(\widehat{\text{ATT}}_{dr} - \text{ATT}) &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\varphi_1(O_i, \hat{\eta}^{-k}) - \text{ATT} \right) + \frac{1}{n} \sum_{i=1}^n \varphi_1(O_i, \hat{\eta}^{-k}) \sqrt{n} \left(\frac{\pi}{\hat{\pi}} - 1 \right) \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\varphi_1(O_i, \hat{\eta}^{-k}) - \text{ATT} \right) + \text{ATT} \sqrt{n} \left(\frac{\pi}{\hat{\pi}} - 1 \right) + o_p(1) \\ &= \underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^n (\varphi_1(O_i, \hat{\eta}^{-k}) - \text{ATT})}_A - \underbrace{\frac{\text{ATT}}{\pi} \cdot \frac{1}{\sqrt{n}} \sum_{i=1}^n (D_i - \pi)}_B + o_p(1). \end{aligned}$$

where the first equality holds by adding and subtracting $\frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi_1(O_i, \hat{\eta}^{-k})$, the second equality holds because $\frac{1}{n} \sum_{i=1}^n \varphi_1(O_i, \hat{\eta}^{-k})$ is consistent for ATT and by the continuous mapping theorem, and the last equality holds by rearranging terms and the continuous mapping theorem. For term

A, notice that

$$\begin{aligned}
\frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\varphi_1(O_i, \hat{\eta}^{-k}) - \text{ATT} \right) &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\varphi_1(O_i, \eta) - \text{ATT} \right) + \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\varphi_1(O_i, \hat{\eta}^{-k}) - \varphi_1(O_i, \eta) \right) \\
&= \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\varphi_1(O_i, \eta) - \text{ATT} \right) + \sqrt{n} \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \mathbb{E} \left[\left(\varphi_1(O_i, \hat{\eta}^{-k}) - \varphi_1(O_i, \eta) \right) \mid \hat{\eta}^{-k} \right] \\
&\quad + \frac{1}{\sqrt{n}} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left\{ \left(\varphi_1(O_i, \hat{\eta}^{-k}) - \varphi_1(O_i, \eta) \right) - \mathbb{E} \left[\left(\varphi_1(O_i, \hat{\eta}^{-k}) - \varphi_1(O_i, \eta) \right) \mid \hat{\eta}^{-k} \right] \right\} \\
&=: T_1 + T_2 + T_3
\end{aligned}$$

where the first equality holds by adding and subtracting $\frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi_1(O_i, \eta)$, and the second equality holds by adding and subtracting $\sqrt{n} \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \mathbb{E} \left[\left(\varphi_1(O_i, \hat{\eta}^{-k}) - \varphi_1(O_i, \eta) \right) \mid \hat{\eta}^{-k} \right]$.

We show in Lemma S3 that $T_2 = o_p(1)$, and we show in Lemma S4 that $T_3 = o_p(1)$. Thus, combining the results above, we have that

$$\sqrt{n}(\widehat{\text{ATT}}_{dr} - \text{ATT}) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi(O_i) + o_p(1) \xrightarrow{d} \mathcal{N}(0, V_{dr})$$

with $\varphi(O_i, \eta)$ as defined in the proposition. □

Lemma S3. *Under Assumptions 1, 2, 6, 7 and 9, and S2,*

$$\sqrt{n} \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \mathbb{E} \left[\left(\varphi_1(O_i, \hat{\eta}^{-k}) - \varphi_1(O_i, \eta) \right) \mid \hat{\eta}^{-k} \right] = o_p(1).$$

Proof. Define $T_2(k) := \varphi_1(O, \hat{\eta}^{-k}) - \varphi_1(O, \eta)$, so that

$$\sqrt{n} \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \mathbb{E} \left[\left(\varphi_1(O_i, \hat{\eta}^{-k}) - \varphi_1(O_i, \eta) \right) \mid \hat{\eta}^{-k} \right] = \sqrt{n} \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \mathbb{E} [T_2(k) \mid \hat{\eta}^{-k}],$$

For a particular fold k , it holds that

$$\begin{aligned}
\mathbb{E} [T_2(k) \mid \hat{\eta}^{-k}] &= \mathbb{E} \left[-\frac{D}{\pi} \hat{\nu}_0^{-k} - \frac{(1-D)}{\pi} (\hat{m}_0^{-k} - \hat{\nu}_0^{-k}) \frac{\hat{p}^{-k}}{1 - \hat{p}^{-k}} - \frac{(1-D)}{\pi} (\Delta Y_{t^*} - \hat{m}_0^{-k}) \hat{\omega}_0^{-k} \mid \hat{\eta}^{-k} \right] \\
&\quad + \mathbb{E} \left[\frac{D}{\pi} \nu_0 + \frac{(1-D)}{\pi} (m_0 - \nu_0) \frac{p}{1-p} + \frac{(1-D)}{\pi} (\Delta Y_{t^*} - m_0) \omega_0 \mid \hat{\eta}^{-k} \right] \\
&= - \underbrace{\mathbb{E} \left[\frac{D}{\pi} (\hat{\nu}_0^{-k} - \nu_0) \mid \hat{\eta}^{-k} \right]}_A \\
&\quad - \underbrace{\mathbb{E} \left[\frac{1-D}{\pi} (\hat{m}_0^{-k} - m_0) \frac{\hat{p}^{-k}}{1 - \hat{p}^{-k}} \mid \hat{\eta}^{-k} \right]}_B - \underbrace{\mathbb{E} \left[\frac{1-D}{\pi} m_0 \left(\frac{\hat{p}^{-k}}{1 - \hat{p}^{-k}} - \frac{p}{1-p} \right) \mid \hat{\eta}^{-k} \right]}_C
\end{aligned}$$

$$\begin{aligned}
& + \underbrace{\mathbb{E} \left[\frac{1-D}{\pi} (\hat{\nu}_0^{-k} - \nu_0) \frac{\hat{p}^{-k}}{1 - \hat{p}^{-k}} \mid \eta^{-k} \right]}_D + \underbrace{\mathbb{E} \left[\frac{1-D}{\pi} \nu_0 \left(\frac{\hat{p}^{-k}}{1 - \hat{p}^{-k}} - \frac{p}{1-p} \right) \mid \eta^{-k} \right]}_E \\
& + \underbrace{\mathbb{E} \left[\frac{(1-D)}{\pi} (\hat{m}_0^{-k} - m_0) \hat{\omega}_0^{-k} \mid \eta^{-k} \right]}_F - \underbrace{\mathbb{E} \left[\frac{(1-D)}{\pi} (\Delta Y_{t^*} - m_0) (\hat{\omega}_0^{-k} - \omega_0) \mid \eta^{-k} \right]}_G
\end{aligned}$$

which the first equality holds by the definition of $T_2(k)$, and the second equality holds by adding and subtracting terms and rearranging. Terms C and E cancel by the law of iterated expectations. Term $G = 0$ by the law of iterated expectations.

$$\begin{aligned}
-A + D &= -\mathbb{E} \left[(\hat{\nu}_0^{-k} - \nu_0) \mid \hat{\eta}^{-k}, D = 1 \right] + \frac{1-\pi}{\pi} \mathbb{E} \left[(\hat{\nu}_0^{-k} - \nu_0) \frac{\hat{p}^{-k}}{1 - \hat{p}^{-k}} \mid \eta^{-k}, D = 0 \right] \\
&= -\frac{1-\pi}{\pi} \mathbb{E} \left[(\hat{\nu}_0^{-k} - \nu_0) \frac{p}{1-p} \mid \eta^{-k}, D = 0 \right] + \frac{1-\pi}{\pi} \mathbb{E} \left[(\hat{\nu}_0^{-k} - \nu_0) \frac{\hat{p}^{-k}}{1 - \hat{p}^{-k}} \mid \eta^{-k}, D = 0 \right] \\
&= \frac{1-\pi}{\pi} \mathbb{E} \left[\frac{(\hat{\nu}_0^{-k} - \nu_0)(\hat{p}^{-k} - p)}{(1-p)(1 - \hat{p}^{-k})} \mid \eta^{-k}, D = 0 \right]
\end{aligned}$$

where the first equality holds by the definitions of A and D , the second equality holds by a change of measure argument, and the third equality holds by combining terms and canceling. Next,

$$\begin{aligned}
-B + F &= -\frac{1-\pi}{\pi} \mathbb{E} \left[(\hat{m}_0^{-k} - m_0) \frac{\hat{p}^{-k}}{1 - \hat{p}^{-k}} \mid \eta^{-k}, D = 0 \right] + \frac{1-\pi}{\pi} \mathbb{E} \left[(\hat{m}_0^{-k} - m_0) \hat{\omega}_0^{-k} \mid \eta^{-k}, D = 0 \right] \\
&= -\frac{1-\pi}{\pi} \mathbb{E} \left[(\hat{m}_0^{-k} - m_0) \left(\frac{\hat{p}^{-k}}{1 - \hat{p}^{-k}} - \frac{p}{1-p} \right) \mid \eta^{-k}, D = 0 \right] \\
&\quad + \frac{1-\pi}{\pi} \mathbb{E} \left[(\hat{m}_0^{-k} - m_0) (\hat{\omega}_0^{-k} - \omega_0) \mid \eta^{-k}, D = 0 \right] \\
&= -\frac{1-\pi}{\pi} \mathbb{E} \left[\frac{(\hat{m}_0^{-k} - m_0)(\hat{p}^{-k} - p)}{(1-p)(1 - \hat{p}^{-k})} \mid \eta^{-k}, D = 0 \right] \\
&\quad + \frac{1-\pi}{\pi} \mathbb{E} \left[(\hat{m}_0^{-k} - m_0) (\hat{\omega}_0^{-k} - \omega_0) \mid \eta^{-k}, D = 0 \right]
\end{aligned}$$

where the first equality holds by the law of iterated expectations, the second equality holds by adding and subtracting $\frac{1-\pi}{\pi} \mathbb{E} \left[(\hat{m}_0^{-k} - m_0) \frac{p}{1-p} \mid \eta^{-k}, D = 0 \right]$ and by applying the law of iterated expectations and the definition of ω_0 in the second term, and the last equality holds by rearranging the first term.

The previous results imply that

$$\begin{aligned}
\mathbb{E} [T_2(k) \mid \hat{\eta}^{-k}] &= \frac{1-\pi}{\pi} \mathbb{E} \left[\frac{(\hat{\nu}_0^{-k} - \nu_0)(\hat{p}^{-k} - p)}{(1-p)(1 - \hat{p}^{-k})} \mid \eta^{-k}, D = 0 \right] \\
&\quad - \frac{1-\pi}{\pi} \mathbb{E} \left[\frac{(\hat{m}_0^{-k} - m_0)(\hat{p}^{-k} - p)}{(1-p)(1 - \hat{p}^{-k})} \mid \eta^{-k}, D = 0 \right]
\end{aligned}$$

$$\begin{aligned}
& + \frac{1-\pi}{\pi} \mathbb{E} \left[\left(\hat{m}_0^{-k} - m_0 \right) \left(\hat{\omega}_0^{-k} - \omega_0 \right) \mid \eta^{-k}, D = 0 \right] \\
& \leq C \left(\|\hat{\nu}_0^{-k} - \nu_0\|_2 \times \|\hat{p}^{-k} - p\|_2 + \|\hat{m}_0^{-k} - m_0\|_2 \times \|\hat{p}^{-k} - p\|_2 + \|\hat{m}_0^{-k} - m_0\|_2 \times \|\hat{\omega}_0^{-k} - \omega_0\|_2 \right) \\
& = o_p(n^{-1/2})
\end{aligned}$$

where the first equality holds by combining the previous results, the inequality on the fourth line holds by Assumptions S2 and 7 and by the Cauchy-Schwarz inequality, and the last line holds by Assumption 9.

Since this result holds for any fold k and $|\mathcal{I}_k|$ is the same order as n , it implies the claim in the lemma. $\square$

Lemma S4. *Under Assumptions 1, 2, 6, 7 and 9, and S2,*

$$\frac{1}{\sqrt{n}} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left\{ \left(\varphi_1(O_i, \hat{\eta}^{-k}) - \varphi_1(O_i, \eta) \right) - \mathbb{E} \left[\left(\varphi_1(O_i, \hat{\eta}^{-k}) - \varphi_1(O_i, \eta) \right) \mid \hat{\eta}^{-k} \right] \right\} = o_p(1)$$

Proof. Define $\dot{\varphi}_{1,i}(k) := \varphi_1(O_i, \hat{\eta}^{-k}) - \varphi_1(O_i, \eta) - \mathbb{E} \left[\left(\varphi_1(O_i, \hat{\eta}^{-k}) - \varphi_1(O_i, \eta) \right) \mid \hat{\eta}^{-k} \right]$, so that

$$\frac{1}{\sqrt{n}} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left\{ \left(\varphi_1(O_i, \hat{\eta}^{-k}) - \varphi_1(O_i, \eta) \right) - \mathbb{E} \left[\left(\varphi_1(O_i, \hat{\eta}^{-k}) - \varphi_1(O_i, \eta) \right) \mid \hat{\eta}^{-k} \right] \right\} = \sum_{k=1}^K T_3(k)$$

where

$$T_3(k) := \frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_k} \dot{\varphi}_{1,i}(k).$$

For any fold k , notice that $\mathbb{E}[\dot{\varphi}_1(k) \mid \hat{\eta}^{-k}] = 0$, which holds immediately, and implies that

$$\mathbb{E} [T_3(k) \mid \hat{\eta}^{-k}] = 0.$$

Further, notice that

$$\begin{aligned}
\text{Var}(\dot{\varphi}_1(k) \mid \hat{\eta}^{-k}) & \leq \mathbb{E} \left[\left(\varphi_1(O; \hat{\eta}^{-k}) - \varphi_1(O; \eta) \right)^2 \mid \hat{\eta}^{-k} \right] \\
& \leq C \left(\|\hat{m}_0 - m_0\|_2^2 + \|\hat{\nu}_0 - \nu_0\|_2^2 + \|\hat{p} - p\|_2^2 + \|\hat{\omega}_0 - \omega_0\|_2^2 \right) \\
& = o_p(1)
\end{aligned} \tag{S14}$$

where the first line holds by the definition of variance, the second line holds by adding and subtracting terms and then applying the Cauchy-Schwarz inequality under Assumption S2(iii)-(v), and the last line holds because the nuisance functions are L^2 consistent by Assumption S2(ii).

This implies that

$$\text{Var}(T_3(k) \mid \hat{\eta}^{-k}) = \text{Var} \left(\frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_k} \dot{\varphi}_{1,i}(k) \mid \hat{\eta}^{-k} \right) = \frac{|\mathcal{I}_k|}{n} \text{Var}(\dot{\varphi}_1(k) \mid \hat{\eta}^{-k}) = o_p(1) \quad (\text{S15})$$

which holds by Assumption 1, because $|\mathcal{I}_k|$ is the same order as n , and by Equation (S14).

Now, for any $\epsilon > 0$ and $\delta > 0$, first note that on the event $\text{Var}(T_3(k) \mid \hat{\eta}^{-k}) \leq \delta$, by Chebyshev's inequality we have that

$$\text{P}(|T_3(k)| > \epsilon \mid \hat{\eta}^{-k}) \leq \frac{\text{Var}(T_3(k) \mid \hat{\eta}^{-k})}{\epsilon^2} \leq \frac{\delta}{\epsilon^2}. \quad (\text{S16})$$

Then,

$$\begin{aligned} \text{P}(|T_3(k)| > \epsilon) &= \mathbb{E} [\text{P}(|T_3(k)| > \epsilon \mid \hat{\eta}^{-k})] \\ &\leq \mathbb{E} [\text{P}(|T_3(k)| > \epsilon \mid \hat{\eta}^{-k}) \mathbf{1}\{\text{Var}(T_3 \mid \hat{\eta}^{-k}) \leq \delta\} + \mathbf{1}\{\text{Var}(T_3 \mid \hat{\eta}^{-k}) > \delta\}] \\ &\leq \frac{\delta}{\epsilon^2} \text{P}(\text{Var}(T_3 \mid \hat{\eta}^{-k}) \leq \delta) + \text{P}(\text{Var}(T_3 \mid \hat{\eta}^{-k}) > \delta) \\ &\leq \frac{\delta}{\epsilon^2} + o(1) \end{aligned}$$

where the first equality holds by the law of iterated expectations, the second line holds because the indicator terms sum to one and probabilities are bounded below one, the third line holds by Equation (S16), and the fourth inequality holds by Equation (S15). Since $\delta > 0$ is arbitrary, taking $\delta \rightarrow 0$ implies that $\text{P}(|T_3(k)| > \epsilon) \rightarrow 0$. Applying this argument for all $k = 1, \dots, K$ implies $T_3 = o_p(1)$. $\square$

SB.3 Asymptotic Normality with Staggered Adoption

Appendices SA and SB showed that the imputation and Neyman Orthogonal estimators discussed in the main text are consistent and asymptotically normal in the setting with two time periods. This section discusses how to extend those results to the staggered adoption setting considered in Section 5. We focus on the estimator implied by the identification result in Proposition 2 as we think this is the most likely result to be used in empirical work. Estimating $\text{ATT}(g, t)$ based on this expression is as simple as replacing $(Y_{it^*} - Y_{it^*-1})$ with $(Y_{it} - Y_{ig-1})$, X_{it^*} with X_{it} , and X_{it^*-1} with X_{ig-1} in the preceding estimators. Consistency and asymptotic normality follow from effectively the same arguments as above and involve the same influence function and asymptotic variance up to adjusting the periods, which, following the notation used in the main text, we denote by $\psi_{g,t}(O_i)$ and $\varphi_{g,t}(O_i)$, respectively.

Inference for aggregated parameters like $\text{ATT}^{es}(e)$ and ATT^o , which were discussed in Remark 4, is also straightforward. We only provide a sketch of these arguments as they follow

effectively immediately from existing results in Callaway and Sant'Anna (2021). Notice that all of these parameters can be expressed as

$$\theta = \sum_{g \in \bar{\mathcal{G}}} \sum_{t=g}^T w_{\theta}(g, t) \text{ATT}(g, t),$$

where θ generically denotes some aggregated parameter that depends on the weighting function $w_{\theta}(g, t)$. For example, for $\text{ATT}^{es}(e)$, the weights are given by $\mathbf{1}\{t = g + e\}P(G = g \mid G + e \leq T)$. For ATT^o , the weights are given by $P(G = g \mid G \in \bar{\mathcal{G}})/(T - g + 1)$. One additional challenge in this case is that the weights generally need to be estimated. Thus, an estimator for an aggregated parameter is given by

$$\hat{\theta} = \sum_{g \in \bar{\mathcal{G}}} \sum_{t=g}^T \hat{w}_{\theta}(g, t) \widehat{\text{ATT}}(g, t).$$

We assume that, for all $g \in \bar{\mathcal{G}}$ and $t = g, \dots, T$,

$$\sqrt{n}(\hat{w}_{\theta}(g, t) - w_{\theta}(g, t)) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_{g,t}^w(O_i) + o_p(1),$$

where $\mathbb{E}[\xi_{g,t}^w] = 0$ and $\text{Var}(\xi_{g,t}^w)$ is positive definite and finite. This is a mild high-level assumption about estimating the weights that is satisfied for all the weights considered in Callaway and Sant'Anna (2021) under mild regularity conditions. Under this condition, it follows that

$$\sqrt{n}(\hat{\theta} - \theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_{\theta}(O_i) + o_p(1) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\xi_{\theta}(O)^2])$$

where

$$\xi_{\theta}(O) := \sum_{g \in \bar{\mathcal{G}}} \sum_{t=g}^T \left(w_{\theta}(g, t) \psi_{g,t}(O) + \xi_{g,t}^w(O) \text{ATT}(g, t) \right)$$

The expression above is for the influence function from the imputation estimator, but the same argument applies for the Neyman Orthogonal estimator with $\varphi_{g,t}$ replacing $\psi_{g,t}$.

SC Identification under Parallel Trends for the Bad Control

In this section, we consider identification of the ATT under a parallel trends assumption for the bad control. We focus on the case with two periods, as the extension to staggered treatment

adoption follows along the same lines as the arguments in Section 5 in the main text. Consider the following assumption.

Assumption S3 (Bad Control Parallel Trends).

$$\mathbb{E}[\Delta X_{t^*}(0) \mid W, Z, D = 1] = \mathbb{E}[\Delta X_{t^*}(0) \mid W, Z, D = 0]$$

Under Assumption S3, it is straightforward to see that the average effect of the treatment on the treated of the bad control is identified. In particular, it follows from standard difference-in-differences arguments that

$$\text{ATT}_X := \mathbb{E}[X_{t^*}(1) - X_{t^*}(0) \mid D = 1] = \mathbb{E}[\Delta X_{t^*} \mid D = 1] - \mathbb{E}\left[\mathbb{E}[\Delta X_{t^*} \mid W, Z, D = 0] \mid D = 1\right]. \quad (\text{S17})$$

However, identifying ATT_X is not sufficient for identifying ATT as it depends on the entire distribution of $X_{t^*}(0)$ for the treated group (see Equation (3) in the main text). That being said, what we show next is that if we additionally impose linearity on the parallel trends assumption in Assumption 2, then we can recover the ATT under Assumption S3. Consider the following assumption:

Assumption S4 (Linear Conditional Expectation of $\Delta Y_{t^*}(0)$).

$$\mathbb{E}[\Delta Y_{t^*}(0) \mid X_{t^*}(0), X_{t^*-1}, Z] = X_{t^*}(0)\beta_1 + X_{t^*-1}\beta_2 + Z'\beta_3$$

$(\beta_1, \beta_2, \beta_3)$ are identified from the regression of ΔY_{t^*} on X_{t^*} , X_{t^*-1} , and Z among the untreated group. This is the same assumption as we used for the imputation estimator discussed in the main text. Then, we have the following identification result:

Proposition S1. *Under Assumptions 1, 2, 7, S3, and S4,*

$$\begin{aligned} \text{ATT} = \mathbb{E}[\Delta Y_{t^*} \mid D = 1] - & \left(\mathbb{E}\left[\mathbb{E}[\Delta X_{t^*} \mid W, Z, D = 0] \mid D = 1\right]\beta_1 \right. \\ & \left. + \mathbb{E}[X_{t^*-1} \mid D = 1](\beta_2 + \beta_1) + \mathbb{E}[Z \mid D = 1]'\beta_3 \right), \end{aligned}$$

which is identified.

Proof. Notice that

$$\begin{aligned} \mathbb{E}[\Delta Y_{t^*}(0) \mid D = 1] &= \mathbb{E}\left[\mathbb{E}[\Delta Y_{t^*}(0) \mid X_{t^*}(0), X_{t^*-1}, Z, D = 0] \mid D = 1\right] \\ &= \mathbb{E}\left[X_{t^*}(0)\beta_1 + X_{t^*-1}\beta_2 + Z'\beta_3 \mid D = 1\right] \\ &= \mathbb{E}\left[\Delta X_{t^*}(0) \mid D = 1\right]\beta_1 + \mathbb{E}[X_{t^*-1} \mid D = 1](\beta_2 + \beta_1) + \mathbb{E}[Z \mid D = 1]'\beta_3 \\ &= \mathbb{E}\left[\mathbb{E}[\Delta X_{t^*} \mid W, Z, D = 0] \mid D = 1\right]\beta_1 + \mathbb{E}[X_{t^*-1} \mid D = 1](\beta_2 + \beta_1) + \mathbb{E}[Z \mid D = 1]'\beta_3 \end{aligned}$$

where the first equality holds by Assumption 2 and the same argument used to prove Theorems 1 and 2, the second equality holds by Assumption S4, the third equality holds by adding and subtracting $\mathbb{E}[X_{t^*-1} \mid D = 1]\beta_1$ and rearranging, and the last equality holds by the same argument as in Equation (S17). Plugging this expression into the definition of ATT implies the result. $\square$

To conclude this section, we add one more linearity assumption and derive a corollary of Proposition S1.

Assumption S5 (Linear Conditional Expectation of $\Delta X_{t^*}(0)$).

$$\mathbb{E}[\Delta X_{t^*}(0) \mid W, Z] = W'\delta_1 + Z'\delta_2$$

Notice that, under Assumption S3, (δ_1, δ_2) are identified from the regression of ΔX_{t^*} on W, Z using the untreated group.

Corollary S1. *Under Assumptions 1, 2, 7, S3, S4, and S5,*

$$\begin{aligned} ATT = \mathbb{E}[\Delta Y_{t^*} \mid D = 1] - & \left(\mathbb{E}[W \mid D = 1]'\delta_1\beta_1 + \mathbb{E}[X_{t^*-1} \mid D = 1](\beta_2 + \beta_1) \right. \\ & \left. + \mathbb{E}[Z \mid D = 1]'(\delta_2\beta_1 + \beta_3) \right), \end{aligned}$$

which is identified.

Proof. The result holds immediately by plugging the expression in Assumption S5 into the result of Proposition S1. $\square$

The expression in Corollary S1 suggests a straightforward imputation estimator of the ATT. In a first step, estimate $(\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3)$ and $(\hat{\delta}_1, \hat{\delta}_2)$ from regressing ΔY_{t^*} on (X_{t^*}, X_{t^*-1}, Z) and ΔX_{t^*} on (W, Z) , both using the untreated group. Then, estimate

$$\widehat{ATT} = \frac{1}{n} \sum_{i=1}^n \frac{D_i}{\hat{\pi}} \left(\Delta Y_{i,t^*} - W_i'\hat{\delta}_1\hat{\beta}_1 - X_{i,t^*-1}(\hat{\beta}_2 + \hat{\beta}_1) - Z_i'(\hat{\delta}_2\hat{\beta}_1 + \hat{\beta}_3) \right)$$

SD Monte Carlo Simulations

In this section, we conduct a Monte Carlo study to evaluate the finite-sample performance of the estimators discussed in the main text. We focus on the setting with two time periods and consider five different data generating processes (DGPs) below. The DGPs mainly differ in terms of the model for the bad control, so we start by describing their common structure first. Let $Z_i \sim \mathcal{N}(0, 1)$ and $\eta_i \sim \mathcal{N}(0, 1)$ be mutually independent, with η_i an unobserved heterogeneity term, and let $\varepsilon_i^W, \varepsilon_i^D, \varepsilon_i^{X_1}, \varepsilon_i^{X_2}, \varepsilon_{i1}^Y, \varepsilon_{i2}^Y \sim \mathcal{N}(0, 1)$ be mutually independent noise terms, independent of (Z_i, η_i) . From these primitives, we construct the pre-treatment covariate $W_i = 0.8\eta_i + 0.3Z_i + 0.2\varepsilon_i^W$. The

treatment is assigned as

$$D_i = \mathbf{1}\{0.2Z_i + 0.4W_i + 0.3\eta_i + \varepsilon_i^D > 0\}.$$

The pre-treatment bad control is $X_{i1} = 0.5\eta_i + 0.4Z_i + 0.3\varepsilon_i^{X_1}$, and treatment shifts the bad control by $\lambda = 0.5$ units, so that the treated potential covariate is $X_{i2}(1) = X_{i2}(0) + \lambda$. The untreated potential outcome $Y_{it}(0)$ is linear in $X_{it}(0)$ across all DGPs,

$$Y_{it}(0) = \theta_t + 0.5\eta_i + 0.3Z_i + X_{it}(0) + 0.3\varepsilon_{it}^Y,$$

with $\theta_1 = 0.3$ and $\theta_2 = 0.6$. We set the treated potential outcome at $t = t^*$ to be $Y_{it^*}(1) = Y_{it^*}(0) + (X_{it^*}(1) - X_{it^*}(0)) + \delta$, where δ is the direct effect of the treatment, which we set at $\delta = 0.5$. Thus, the true ATT = $\delta + \lambda = 1.00$. Next, we describe the differences across DGPs.

DGP 1 --- Covariate Unconfoundedness, Linear W

$$X_{i2}(0) = 0.7X_{i1} + 0.3Z_i + 0.2W_i + 0.15 + 0.3\varepsilon_i^{X_2}.$$

DGP 2 --- Covariate Unconfoundedness, Nonlinear W

$$X_{i2}(0) = 0.7X_{i1} + 0.3Z_i + 0.2W_i + 0.03W_i^2 + 0.15 + 0.3\varepsilon_i^{X_2}.$$

DGP 3 --- Simple Covariate Unconfoundedness, Nonlinear X

$$X_{i2}(0) = 0.7X_{i1} + 0.3Z_i + 0.4X_{i1}Z_i + 0.2X_{i1}^2 + 0.15 + 0.3\varepsilon_i^{X_2}.$$

DGP 4 --- Bad Control Parallel Trends

$$X_{i2}(0) = X_{i1} + 0.3Z_i + 0.2W_i + 0.15 + 0.3\varepsilon_i^{X_2}.$$

DGP 5 --- Covariate Unconfoundedness, Nonlinear W, X/W Interaction

$$X_{i2}(0) = 0.7X_{i1} + 0.3Z_i + 0.2W_i + 0.03W_i^2 + 0.05X_{i1}W_i + 0.15 + 0.3\varepsilon_i^{X_2}.$$

We use seven of the nine estimators that were discussed in detail in the application in the main text: TWFE: include BC, TWFE: exclude BC, PT for X, ML: pre-treatment, Imputation, DR, ML, and exclude Imp. include BC and Imp. exclude BC for conciseness. Table S1 summarizes which estimators are consistent for which DGPs.

Even for estimators that are not generally consistent for a given DGP, it is still interesting to check their performance. For example, the functional form assumptions for the imputation and

Table S1: Theoretical Consistency of Each Estimator, by DGP

Method	dgp1	dgp2	dgp3	dgp4	dgp5
TWFE Include BC	×	×	×	×	×
TWFE Exclude BC	×	×	×	×	×
PT for X	×	×	×	✓	×
ML (Pre-treatment)	×	×	✓	×	×
Imputation	✓	×	×	✓	×
DR	✓	×	×	✓	×
ML	✓	✓	✓	✓	✓

doubly robust estimator make them theoretically not consistent for the nonlinear DGPs, but they could still potentially have comparable or better performance than the machine learning estimator in finite samples. Similarly, the assumptions that rationalize **PT for X** and **ML: pre-treatment** do not hold in most DGPs, though they could still potentially perform relatively well. Finally, it is also interesting to compare the finite sample performance across different consistent estimators in DGPs (e.g., 1 and 4) where several estimators are consistent.

In the results below, for each combination of estimator, sample size, and DGP, we report bias ($\widehat{ATT} - ATT$), standard deviation across replications (SD), root mean squared error (RMSE), the ratio of mean standard error to SD (SE/SD), and empirical coverage of nominal 95% confidence intervals.¹ All results provided below are based on 1000 Monte Carlo simulations. We provide these results in Figure S1 and Tables S2 to S6.

The results are in line with the theoretical properties of different estimators that we have discussed in the paper. Mainly, in line with the discussion in the text, traditional approaches that directly include the bad control can perform extremely poorly (**TWFE: include BC**, all five DGPs) as well as approaches that exclude the bad control but make no other adjustment (**TWFE: exclude BC**, DGPs 3 and 4).² Next, even in DGPs where it is not consistent, **PT for X** performs decently well. The exception to this is in DGP 3 (and, to some extent DGP 5), where the likely explanation is due to the more severe nonlinearity in those DGPs. Similarly, **ML: pre-treatment**, even though it is inconsistent in four out of five DGPs, performs fairly well across DGPs, at least in terms of having small bias. However, its bias is nonzero in these DGPs, leading to undercoverage especially as the sample size increases. For both **PT for X** and **ML: pre-treatment**, we conjecture that in most realistic applications, the alternative assumptions imposed by these approaches are likely to, in some sense, be closer to holding than assumptions that lead to fully including or excluding the bad control. Finally, among our estimators that are based on covariate unconfoundedness

¹An SE/SD ratio near 1.0 indicates that the analytical standard errors are well calibrated, and coverage near 0.95 indicates that the confidence intervals have correct size.

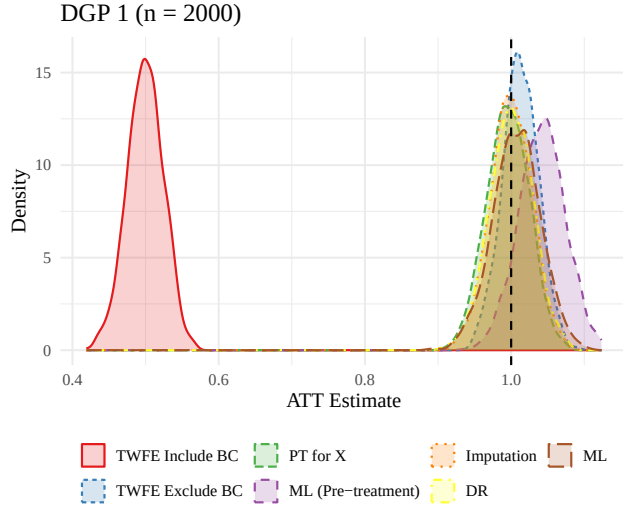
²That excluding the bad control performs fairly well in some DGPs is an artifact of the parameters of these DGPs, and it would be straightforward to make this approach perform poorly by simply changing the parameters in DGPs 1, 2, and 5. It is also noteworthy that this approach is biased in all five DGPs, which results in coverage rates shrinking in the sample size.

(**Imputation**, **DR**, and **ML**), all of them perform decently well across DGPs. The main exception to this is **Imputation** in DGP 3 (and, to some extent, DGP 5), where it performs somewhat worse due to the more severe nonlinearity in these DGPs. Notably, however, even in the DGPs where its functional form assumptions do not hold, the **DR** estimator performs almost on par with or sometimes slightly better than the **ML** estimator. Finally, for some DGPs, **ML**, which is consistent for all of the DGPs considered here, sometimes undercovers. This is driven by standard errors that are sometimes too small relative to the standard deviation of the estimator.

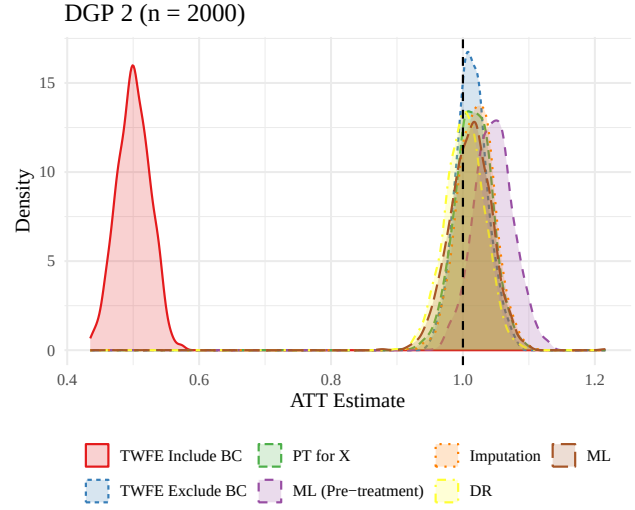
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Callaway, Brantly and Pedro HC Sant’Anna (2021). “Difference-in-differences with multiple time periods”. *Journal of Econometrics* 225.2, pp. 200–230.

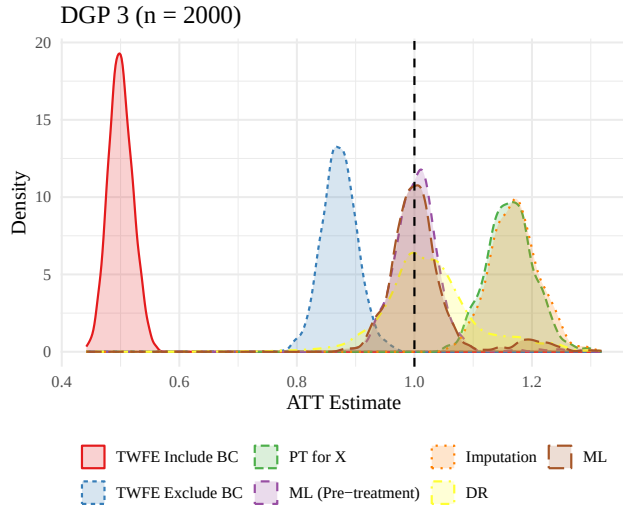
Figure S1: Monte Carlo estimate densities across DGPs



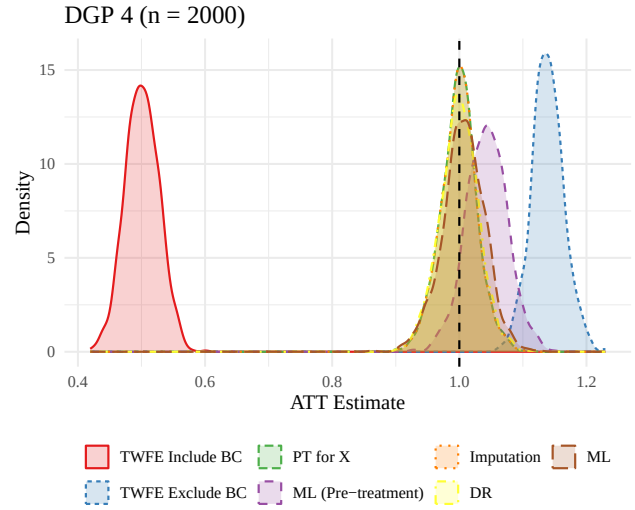
(a) DGP 1



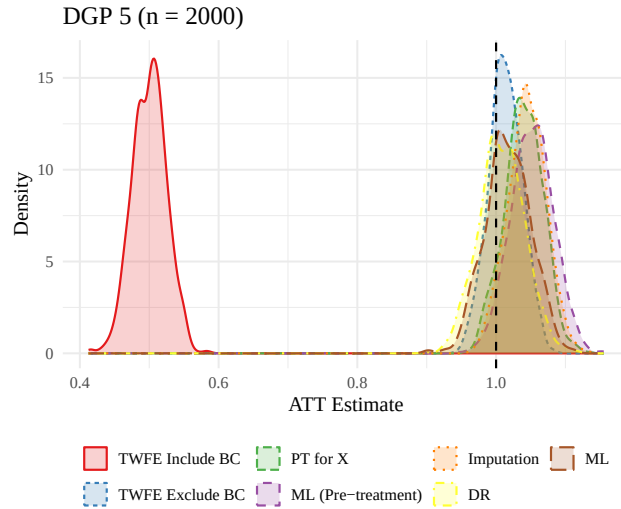
(b) DGP 2



(c) DGP 3



(d) DGP 4



(e) DGP 5

Notes: The figures provides kernel density estimates of ATT across 1,000 replications at $n = 2,000$, for the four main DGPs. The dashed vertical line marks the true ATT = 1.00.

Table S2: Monte Carlo Results: DGP 1

Method	n	Bias	SD	RMSE	SE/SD	Coverage
TWFE Include BC	500	-0.500	0.051	0.502	0.97	0.000
TWFE Include BC	1000	-0.499	0.035	0.501	0.98	0.000
TWFE Include BC	2000	-0.501	0.025	0.501	1.00	0.000
TWFE Exclude BC	500	0.015	0.049	0.051	0.99	0.932
TWFE Exclude BC	1000	0.013	0.036	0.038	0.95	0.924
TWFE Exclude BC	2000	0.013	0.025	0.028	0.99	0.918
PT for X	500	-0.002	0.057	0.057	1.00	0.954
PT for X	1000	-0.004	0.042	0.042	0.97	0.948
PT for X	2000	-0.005	0.029	0.029	0.99	0.945
ML (Pre-treatment)	500	0.031	0.059	0.067	1.01	0.917
ML (Pre-treatment)	1000	0.036	0.043	0.057	1.00	0.870
ML (Pre-treatment)	2000	0.042	0.031	0.052	1.02	0.749
Imputation	500	0.002	0.056	0.056	1.00	0.954
Imputation	1000	-0.000	0.041	0.041	0.96	0.947
Imputation	2000	-0.000	0.028	0.028	1.00	0.947
DR	500	0.003	0.062	0.062	0.97	0.948
DR	1000	-0.000	0.043	0.043	0.96	0.941
DR	2000	0.000	0.029	0.029	0.99	0.948
ML	500	0.011	0.060	0.061	0.99	0.935
ML	1000	0.007	0.045	0.045	0.97	0.940
ML	2000	0.007	0.032	0.033	0.97	0.937

Notes: The table provides Monte Carlo results for DGP 1. The rows are organized by the seven estimators discussed in the main text. Results are based on 1000 Monte Carlo simulations.

Table S3: Monte Carlo Results: DGP 2

Method	n	Bias	SD	RMSE	SE/SD	Coverage
TWFE Include BC	500	-0.499	0.050	0.501	0.99	0.000
TWFE Include BC	1000	-0.501	0.035	0.503	1.00	0.000
TWFE Include BC	2000	-0.499	0.025	0.500	0.99	0.000
TWFE Exclude BC	500	0.013	0.048	0.050	1.00	0.941
TWFE Exclude BC	1000	0.012	0.034	0.036	1.01	0.939
TWFE Exclude BC	2000	0.013	0.024	0.027	1.02	0.910
PT for X	500	0.016	0.059	0.061	0.96	0.929
PT for X	1000	0.015	0.040	0.042	1.02	0.940
PT for X	2000	0.016	0.028	0.032	1.02	0.922
ML (Pre-treatment)	500	0.036	0.061	0.071	0.98	0.902
ML (Pre-treatment)	1000	0.041	0.041	0.058	1.06	0.859
ML (Pre-treatment)	2000	0.046	0.030	0.055	1.07	0.706
Imputation	500	0.021	0.058	0.062	0.97	0.925
Imputation	1000	0.019	0.039	0.043	1.02	0.931
Imputation	2000	0.020	0.028	0.034	1.02	0.901
DR	500	0.002	0.063	0.063	0.97	0.938
DR	1000	0.002	0.042	0.042	1.00	0.951
DR	2000	0.003	0.029	0.030	0.99	0.952
ML	500	0.016	0.061	0.063	0.98	0.931
ML	1000	0.014	0.042	0.044	1.03	0.940
ML	2000	0.011	0.032	0.034	0.99	0.939

Notes: The table provides Monte Carlo results for DGP 2. The rows are organized by the seven estimators discussed in the main text. Results are based on 1000 Monte Carlo simulations.

Table S4: Monte Carlo Results: DGP 3

Method	n	Bias	SD	RMSE	SE/SD	Coverage
TWFE Include BC	500	-0.502	0.040	0.504	1.02	0.000
TWFE Include BC	1000	-0.499	0.029	0.500	1.01	0.000
TWFE Include BC	2000	-0.500	0.020	0.500	1.00	0.000
TWFE Exclude BC	500	-0.128	0.060	0.141	1.00	0.409
TWFE Exclude BC	1000	-0.128	0.041	0.134	1.02	0.143
TWFE Exclude BC	2000	-0.127	0.030	0.131	1.00	0.017
PT for X	500	0.165	0.084	0.185	0.97	0.474
PT for X	1000	0.163	0.057	0.173	1.02	0.193
PT for X	2000	0.163	0.040	0.168	1.01	0.013
ML (Pre-treatment)	500	0.009	0.067	0.068	1.08	0.961
ML (Pre-treatment)	1000	0.007	0.046	0.047	1.11	0.957
ML (Pre-treatment)	2000	0.005	0.043	0.044	0.85	0.952
Imputation	500	0.172	0.084	0.192	0.99	0.445
Imputation	1000	0.171	0.057	0.180	1.02	0.169
Imputation	2000	0.170	0.042	0.175	1.00	0.009
DR	500	0.004	0.157	0.157	0.81	0.896
DR	1000	0.013	0.103	0.104	0.87	0.889
DR	2000	0.027	0.080	0.085	0.76	0.845
ML	500	0.010	0.070	0.070	1.04	0.956
ML	1000	0.008	0.052	0.053	0.99	0.951
ML	2000	0.013	0.063	0.064	0.60	0.892

Notes: The table provides Monte Carlo results for DGP 3. The rows are organized by the seven estimators discussed in the main text. Results are based on 1000 Monte Carlo simulations.

Table S5: Monte Carlo Results: DGP 4

Method	n	Bias	SD	RMSE	SE/SD	Coverage
TWFE Include BC	500	-0.500	0.051	0.503	1.04	0.000
TWFE Include BC	1000	-0.501	0.039	0.503	0.96	0.000
TWFE Include BC	2000	-0.499	0.026	0.500	1.00	0.000
TWFE Exclude BC	500	0.140	0.048	0.148	1.02	0.194
TWFE Exclude BC	1000	0.140	0.036	0.144	0.97	0.022
TWFE Exclude BC	2000	0.139	0.025	0.141	0.98	0.000
PT for X	500	0.000	0.054	0.054	1.04	0.956
PT for X	1000	-0.001	0.040	0.040	1.00	0.949
PT for X	2000	-0.001	0.029	0.029	0.96	0.933
ML (Pre-treatment)	500	0.043	0.057	0.071	1.07	0.910
ML (Pre-treatment)	1000	0.042	0.042	0.060	1.05	0.846
ML (Pre-treatment)	2000	0.043	0.032	0.054	1.00	0.727
Imputation	500	0.000	0.054	0.054	1.05	0.953
Imputation	1000	-0.001	0.040	0.040	1.00	0.949
Imputation	2000	-0.001	0.029	0.029	0.96	0.934
DR	500	0.001	0.058	0.058	1.03	0.951
DR	1000	-0.001	0.041	0.041	1.00	0.946
DR	2000	-0.000	0.030	0.030	0.95	0.929
ML	500	0.021	0.058	0.061	1.06	0.945
ML	1000	0.013	0.043	0.045	1.02	0.937
ML	2000	0.006	0.033	0.034	0.95	0.935

Notes: The table provides Monte Carlo results for DGP 4. The rows are organized by the seven estimators discussed in the main text. Results are based on 1000 Monte Carlo simulations.

Table S6: Monte Carlo Results: DGP 5

Method	n	Bias	SD	RMSE	SE/SD	Coverage
TWFE Include BC	500	-0.503	0.050	0.506	0.97	0.000
TWFE Include BC	1000	-0.499	0.036	0.501	0.96	0.000
TWFE Include BC	2000	-0.500	0.025	0.500	1.00	0.000
TWFE Exclude BC	500	0.011	0.048	0.050	1.00	0.944
TWFE Exclude BC	1000	0.013	0.037	0.039	0.94	0.923
TWFE Exclude BC	2000	0.013	0.024	0.027	1.03	0.918
PT for X	500	0.037	0.057	0.068	1.02	0.901
PT for X	1000	0.041	0.043	0.060	0.94	0.830
PT for X	2000	0.040	0.028	0.049	1.04	0.719
ML (Pre-treatment)	500	0.040	0.059	0.071	1.03	0.901
ML (Pre-treatment)	1000	0.049	0.046	0.067	0.97	0.798
ML (Pre-treatment)	2000	0.052	0.031	0.060	1.05	0.641
Imputation	500	0.042	0.055	0.069	1.03	0.892
Imputation	1000	0.046	0.043	0.063	0.94	0.788
Imputation	2000	0.044	0.027	0.052	1.04	0.646
DR	500	-0.002	0.066	0.066	0.98	0.958
DR	1000	0.005	0.048	0.049	0.93	0.923
DR	2000	0.008	0.033	0.034	0.95	0.927
ML	500	0.020	0.059	0.062	1.01	0.937
ML	1000	0.020	0.046	0.050	0.95	0.911
ML	2000	0.017	0.033	0.037	0.97	0.900

Notes: The table provides Monte Carlo results for DGP 5. The rows are organized by the seven estimators discussed in the main text. Results are based on 1000 Monte Carlo simulations.